BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6465

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Yeasts are unicellular:

(A) Algae
(B) Protozoans
(C) Fungi
(D) Bacteria

Q2. Entamoeba histolytica causes:

(A) Dysentery
(B) Fever
(C) Cholera
(D) Hepatitis

Q3. Bacterial pathogenicity is due to:

(A) Capsule
(B) Cell wall
(C) Cyst
(D) Slime

Q4. Bacteriophage replicates only in cells:

(A) Animal
(B) Plant
(C) Bacteria
(D) Fungi

Q5. The number of chromosomes in drosophila melanogaster is:

(A) 08
(B) 14
(C) 16
(D) 26

Q6. Enzymes lower down the energy of:

(A) Kinetic
(B) Activation
(C) Potential
(D) Ionic

Q7. Cotton is the pure form of:

(A) Glycogen
(B) Waxes
(C) Cellulose
(D) Amino acids

Q8. In human body the amount of iron is:

(A) 4%
(B) 0.4%
(C) 0.04%
(D) 0.004%

Q9. Which is not granulocytes?

(A) Monocytes
(B) Neutrophils
(C) Eosinophils
(D) Basophils

Q10. The right atrium receives deoxygenated blood from these parts of the body except:

(A) Brain
(B) Liver
(C) Lungs
(D) Kidneys

Q11. The respiratory system is the most efficient in:

(A) Man
(B) Birds
(C) Reptiles
(D) Fish

Q12. Which animal has no need for a gall bladder?

(A) Cat
(B) Dog
(C) Lion
(D) Goat

Q13. Functional group of chlorophyll 'a' is:

(A) CH_3
(B) CHO
(C) COOH
(D) OH

Q14. Lactic acid is formed in these except during extreme physical activities:

(A) Man
(B) Fish
(C) Horse
(D) Cat

Q15. These are arachnids except:

(A) Scorpion
(B) Beetles
(C) Ticks
(D) Mites

Q16. Which one is not a pseudocoelomate?

(A) Ascaris
(B) Earthworm
(C) Ancylostoma
(D) Round worms

Q17. An ovule is an Integumented indehiscent:

(A) Micro-sporangium
(B) Seed
(C) Sporogonium
(D) Mega-sporangium

Biology (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6465

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) How basic proteins play their role in combined form with nucleic acids?

(ii) Why enzymes need optimum pH for their proper functioning? Give one example.

(iii) Why some enzymatic reactions occur in series to form the final products? Explain briefly.

(iv) Which model for enzyme substrate interaction is more supported? Discuss briefly that model.

(v) Name four diseases caused by fungi.

(vi) How karyogamy is different from plasmogamy?

(vii) Write four characteristics of mammals.

(viii) Give economic importance of mollusca.

(ix) Define the term regeneration and name animal which exhibit it.

(x) What are coral reefs and where they are found?

(xi) Differentiate between chlorophyll-a and chlorophyll-b.

(xii) What are the differences between alcoholic and lactic acid fermentation?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Why organ system is less complex in plants as compared to animals?

(ii) What are adverse effects of use of chemicals during its use to control pests?

(iii) What is difference between cisternae and cristae?

(iv) Why mitochondria are called self-replicating organelle?

(v) From where the Giant Amoeba obtain energy?

(vi) How ciliates differ from protozoans?

(vii) Name the parasitic flagellates and disease caused by it and how it is transmitted?

(viii) Why fungus like protists are not fungi?

(ix) Differentiate between antheridiophore and archegoniophore.

(x) Write names of two extinct and two living members of psilopsida.

(xi) Define guttation. What factors affect it?

(xii) Write role of lymphatic system in defense of body.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Write short note on polio.

(ii) Define plasmids. What is its use?

(iii) Write any two digestive functions of liver.

(iv) What is ulcer? Write few lines on it.

(v) Differentiate ectoparasites and endoparasites with examples.

(vi) What is the function of parabronchi in birds?

(vii) Define photorespiration. Give its consequences.

(viii) Define Calvin Benson Cycle.

(ix) What the role spiracles play in cockroach respiration?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5. (Marks 04) (a)Why is some diseases vaccination is used while for others preventive measures are adopted?

(b) Discuss major solutes and their role in blood plasma.

Q6. (Marks 04) (a)Explain the importance of carbon in living organisms.

(b) Describe different methods of asexual reproduction in fungi.

Q7. (Marks 02,02) (a)How does physical methods differ from chemical methods to control bacteria?

(b) How does evolution of microphyll differ from evolution of megaphyll?

Q8. (Marks 01,03) (a)Write a note on five kingdom classification system of Robert Whittaker.

(b) Sketch the process of glycolysis. (No description required)

Q9. (Marks 04) (a)What is endoplasmic reticulum? Write its types and functions.

(b) Describe digestion of food in small intestine.

GROUP – II

Biology (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6462

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Branch of biology which deals with the study of distribution of animals in nature is called:

(A) Biogeography
(B) Zoogeography
(C) Phytogeography
(D) Palaeontology

Q2. Lipoproteins are structural framework of:

(A) Chromosomes
(B) Nucleoid
(C) Membranes
(D) Chromatin

Q3. During metabolic reactions, these substances can be used again and again:

(A) Co-factor
(B) Enzymes
(C) Enzymes and coenzymes
(D) Prosthetic group

Q4. A lipoprotein membrane (crista) contains F1-Particles and different:

(A) Nucleotides
(B) RNA
(C) Electron carriers
(D) Nucleoli

Q5. CD_4-receptor is present on:

(A) Plasma cells
(B) Helper T-cells
(C) B-Lymphocytes
(D) Natural killer cells

Q6. When a bacterium posses a single flagellum at one pole is termed as:

(A) Atrichous
(B) Lophotrichous
(C) Amphitrichous
(D) Monotrichous

Q7. Entamoeba histolytica causes:

(A) Intestinal parasite
(B) Lung parasite
(C) Brain parasite
(D) Blood parasite

Q8. Reindeer moss is a:

(A) Moss
(B) Animal
(C) Lichen
(D) Plant

Q9. All seed producing plants are called:

(A) Arthrophytes
(B) Angiosperms
(C) Pteridophytes
(D) Spermatophytes

Q10. Sponges are protandrous hermaphrodite because:

(A) Female sex cells develop first
(B) Ovaries develop first
(C) Sperms cells develop first
(D) Testes develop first

Q11. A solid double ventral nerve cord is present in earthworm, hence earthworm is dissected from:

(A) Ventral side
(B) Dorsal side
(C) Lateral side
(D) Ventrolateral

Q12. Pyruvate decarboxylase activity is inhibited by; optimal concentration of:

(A) ATP
(B) NAD^+
(C) NADH
(D) Citrate

Q13. Two molecules of reduces NAD produces ____ molecules of ATP in respiratory chain.

(A) 3
(B) 6
(C) 4
(D) 2

Q14. A neurotic disorder in slightly older girls is called:

(A) Anorexia nervosa
(B) Dyspepsia
(C) Obesity
(D) Bulimia nervosa

Q15. Emphysema is the breakdown of:

(A) Bronchi
(B) Alveoli
(C) Trachea
(D) Bronchioles

Q16. Highest blood pressure is in aorta is generated by contraction of:

(A) Right ventricle
(B) Left atrium
(C) Right atrium
(D) Left ventricle

Q17. During filling of heart chambers, walls of chambers are relaxed indicate:

(A) Atrial diastole
(B) Cardiac diastole
(C) Ventricular systole
(D) Ventricular diastole

Biology (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6462

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Why cellulose is digested in herbivores but not in humans?

(ii) Why pepsin is secreted in form of pepsinogen?

(iii) Name an enzyme which act in acidic medium and also give its pH.

(iv) Why inorganic ions cannot be used as co-enzyme?

(v) Differentiate between ectomycorrhizae and endomycorrhizae.

(vi) What are predator fungi? Give an example.

(vii) Differentiate between diploblastic and triploblastic organisms.

(viii) How coral reefs are formed?

(ix) Write two adaptations of parasites for parasitic mode of life.

(x) Write two harmful effects of insects.

(xi) Write the molecular formulas of chlorophyll a and b.

(xii) What is lactic acid fermentation? Also write its equation.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) What is hydroponic culture technique? Give its uses.

(ii) How and when a hypothesis becomes a theory?

(iii) Do you know about the body in the nucleus which is stained dark?

(iv) Can you compare the process of pinocytosis with phagocytosis?

(v) How the pathogen of malaria completes its life cycle?

(vi) Can you give economic importance of algae?

(vii) Which type of pigments are present in algae?

(viii) Give two characteristics of red algae.

(ix) What is protonema? In which group of bryophyte it is produced?

(x) Give four examples of ferns.

(xi) What is pressure potential?

(xii) Define active immunity.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) What is provirus?

(ii) Differentiate between prokaryotic and eukaryotic cytoplasm.

(iii) What is dyspepsia? Give its two symptoms.

(iv) What is appendicitis?

(v) Define peristalsis.

(vi) What are vocal cords?

(vii) Why is respiration important to living organisms?

(viii) Why myoglobin is termed as muscle haemoglobin?

(ix) Why most of the cetaceans have high concentration of myoglobin in muscles?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5. (Marks 04) (a)How can we protect ourselves from the incurable diseases?

(b) What is transpiration? Explain the stomatal transpiration.

Q6. (Marks 04) (a)Write a note on primary and secondary structure of proteins.

(b) Give economic losses due to fungi.

Q7. (Marks 02,02) (a)There is diversity in respiration of bacteria. Prove it. Give examples.

(b) To, what does alternation of generations refer in plants (Bryophytes)? How it is significant?

Q8. (Marks 01,03) (a)Define viruses and describe their characteristics.

(b) Discuss Krebs cycle with sketch in detail.

Q9. (Marks 04) (a)Differentiate between prokaryotic and eukaryotic cells.

(b) Write a note on absorption of food in ileum of human.